FitLog AI

System Architecture & Technical Specifications

PRODUCTION READY v0.0.1

1. Executive Summary

FitLog AI is a high-performance, full-stack fitness and body metrics tracking ecosystem. Built on a serverless architecture using React and Firebase, it integrates Google's cutting-edge Gemini AI to provide real-time, context-aware coaching. The system is designed for multi-platform delivery, functioning as a high-speed Web App, a PWA, and a native Android application via Capacitor.

2. Technology Stack

Layer	Technology	Purpose
Frontend	React 18.3 + Vite 5.4	Core UI library and optimized build pipeline
Icons	Lucide React	Industry-standard professional SVG icons
Analytics	Chart.js 4	High-performance data visualization
AI Engine	Google Gemini API	Generative AI for personalized fitness coaching
Backend/Auth	Firebase Auth	OIDC/OAuth and secure session management
Database	Firebase Realtime DB	NoSQL real-time JSON data synchronization
Mobile Bridge	Capacitor 8	Cross-platform native Android API bridge

3. System Architecture



Key Architectural Decisions

- **Direct Frontend AI Integration:** Utilizes the `@google/generative-ai` SDK directly on the client to eliminate proxy latency and simplify the infrastructure footprint.
- **Resilient Model Fallback Matrix:** Implements a sequential trial logic (Gemini 2.0 Flash → Gemini Pro) across multiple API versions (v1, v1beta) to ensure maximum uptime across varying API key permissions.
- **Serverless Operations:** The entire production environment operates on zero-base-cost infrastructure (Firebase Free Tier + Google AI Studio), ensuring high profit margins for SaaS deployments.

4. Data Schema (NoSQL JSON Tree)

```
{
  "users": {
    "{userId}": {
      "exercises": { "{exerciseId}": { "name": "Bench Press", "createdAt": 1723... } },
      "records": { "{recordId}": { "exerciseId": "...", "weight": 100, "reps": 5, "date": "..." } },
      "bodyMetrics": { "{metricId}": { "weight": 75, "bodyFat": 12, "date": "..." } }
    }
  }
}
```

5. Professional UI/UX Features

- **Dynamic Theme Engine:** Custom context-driven styling system supporting zero-latency Dark/Light mode transitions.

- **SVG Asset System:** Fully replaced legacy emojis with **Lucide icons**, providing a consistent, high-end professional appearance suitable for corporate or enterprise fitness environments.
- **Export Capabilities:** Client-side generation of `.xlsx` and `.docx` reports using binary-level JS libraries.

6. Project Deliverables

Included in the handover package:

- Optimized React/Vite Source Code
- Fully configured `/android` Capacitor Studio Project
- Firebase Deployment Configuration (Hosting, Rules)
- Gemini AI Integration Pipe & Character Persona Logic

© 2026 FitLog AI Project - Confidential Technical Document